تاریخ:

Haseeb

Haseeb

Lecture #2.

Find the orda and degree g following differential equations.

i) $\frac{d^3y}{dx^3} + \left(\frac{d^2y}{dx^2}\right)^3 + \left(\frac{dy}{dx}\right)^3 + y = 7$

order = 3

degree = 1.

ii) $\frac{d^2y}{dx^2} + 3\left(\frac{dy}{dx}\right)^2 + 4y = 0$

order = 2

degree = 1

iii) $\frac{d^2y}{dx^2} - \frac{2dy}{dx} + 3y = 0$

order = 2

degree = 1.

iv) $\frac{d^3y}{dx^3} - 3\left(\frac{dy}{dx}\right)^0 + 2y = x^2$

order = 3

degree = 1

v) $y' + (y'')^2 = (x+y'')^2$

order = 2

degree = $(x+y'')^2$

$y' + (y'')^2 \quad x^2 + 2xy'' + (y'')^2 \Rightarrow y' = x^2 + 2xy$

$= (y'')^2 = 2.$ degree = 1

vi) $\frac{d^3y}{dx^3} - \left(\frac{dy}{dx}\right)^{1/2} = 0$

order = 3

degree = ~~not defined due to 1/2~~ - 1

$\left(\frac{d^3y}{dx^3}\right)^2 - \left(\frac{dy}{dx}\right)^{1/2 \times 2} = (0)^2$

degree = 2.

vii) $y = 2\left(\frac{dy}{dx}\right)^2 + 4x\frac{dx}{dy}$

$y\frac{dy}{dx} = 2\left(\frac{dy}{dx}\right)^2 + 4x.$

$y\frac{dy}{dx} - 2\left(\frac{dy}{dx}\right)^2 = 4x$

order = 1

degree = 2

$y = 2\left(\frac{dy}{dx}\right)^2 + \frac{dy}{4x\,dx}$

y

$$y - 2\left(\frac{dy}{dx}\right)^2 = 4x\frac{dx}{dy}$$

$$\frac{1}{y - 2\left(\frac{dy}{dx}\right)^2} = \frac{1}{4x}\cdot\frac{dy}{dx}$$

$$4x = \frac{dy}{dx}\left(y - 2\left(\frac{dy}{dx}\right)^2\right)$$

$$4x = y\frac{dx}{dx} - 2\left(\frac{dy}{dx}\right)^3$$

order = 1, degree = 3

Linear Differential Equation.

An ordinary DE (ODE) is said to be linear if

i) The dependent variable y and it's derivative are of degree one.

ii) No product of y and its derivatives appears

e.g- 1) $y'' + y = e^x$
order = 2
degree = 1
linear.

ii) $y'' + \log(x)y' + xy = 0$
order = 2
degree = 1
linear.

(iii) $y' + \frac{1}{y} = 0$
$y' + y^{-1} = 0$
$yy' + 1 = 0$
$y\frac{dy}{dx} + 1 = 0$
non linear.

4) $y' + y^2 = 0$ = non linear due to y^2.

5) $yy' + \sin(x) = 0$ non linear due to yy'

Solution of Differential Equation.

→ $y = y(x)$ is called a solution of ODE on interval $I \subseteq \mathbb{R}$ if it satisfies the equation and it is defined on I.

$\mathbb{R} = (-\infty, +\infty)$

Example :- Show that $y = ce^{-x} + 2$ is the solution of the ODE $y' + y = 2$ on $\mathbb{R}$

$$y' = c\frac{d}{dx}e^{-x} + e^{-x}\frac{d}{dx}c$$

$$= -ce^{-x} + 0$$

$$-ce^{-x} + ce^{-x} + 2 = 2$$

$$2 = 2$$

Types of Solution:-

1) General Solution:-

Any solution that contains arbitrary constan

$$y = x^2 + C_1x + C_2$$

ii) Particular Solution:- If we assign values to arbitrary constants then solution becomes particular solution.

Lecture # 3.

→ Initial Condition and Boundary Condition.

Initial condition:-

Suppose we have a D.E

$$y'' + y = 0 \quad \rightarrow (1) \qquad y(0) = -1,\ y'(0) = 2$$

$y = c_1 \sin x + c_2 \cos x$ → General Solution.

→ $y(x_0) = y_0$ / $y'(x_0) = y_0$

(x_0: starting point of the system.)

when the values of dependent variable and it's derivatives are given only at one point of independent variable then it is called initial condition.

OR.

A value of unknown function given at starting point

For example:- $y'' + y = 0$, $y(0) = 1$, $y'(0) = 2$ } IVP Initial value Problem.

Boundary Condition:-

Suppose we have a D.E:

$$y'' + y = 0, \quad y(0) = 1, \quad y'(1) = 0$$

When the values of dependent variable are given at more than one point of independent variable then it is called boundary condition.

e.g.:- $\frac{d^2T}{dx^2} = 0$; $y(0) = 100$, $y'(1) = 0$ } Boundary Value Problems.

Formation of Differential Equation:-

Let us consider we have the following relation:

$$f(x, y, c_1, c_2, c_3, \ldots, c_n) = 0 \quad \text{— A}$$

where $c_1, c_2, c_3, \ldots, c_n$ are arbitrary constants

To find the D.E differentiate Eq.(1) n-times and eliminate all the arbitrary constants.

Example #01.

$y = (x^3 + c)e^{-3x}$ ⟶ ; where 'c' is an arbitrary constant.

$$\frac{dy}{dx} = \frac{d}{dx}(x^3 + c)e^{-3x}.$$

$$= (x^3 + c)(-3e^{-3x}) + (e^{-3x})(3x^2)$$

$$y' = 3x^2 e^{-3x} + -\{3e^{-3x}(x^3 + c)$$

$$y' = 3(x^2 e^{-3x} - (x^3 + c)e^{-3x}) \quad -②$$

Putting eq (1) in (2)

$$y' = 3(x^2 e^{-3x} - y)$$

$$\frac{dy}{dx} = 3x^2 e^{-3x} - 3y$$

$$\frac{dy}{dx} + 3y = 3x^2 e^{-3x}$$

which is required DE.

Example #02:-

$$ax + \ln|y| = y + b \quad \text{- A-i)}$$

where a and b are arbitrary constants.

$$ax + \ln|y| - b = y$$

$$\frac{d}{dx} ax + \frac{d}{dx} \ln|y| - \frac{d}{dx} b = \frac{dy}{dx}.$$

$$a + \frac{1}{y}\frac{dy}{dx} = \frac{dy}{dx}$$

$$\frac{d^2y}{dx^2} = 0 + \frac{d}{dx}(y^{-1})$$

$$y'' = -1\, y^{-2} \frac{dy}{dx} \cdot \frac{dy}{dx} + \frac{1}{y}\frac{d^2y}{dx^2}$$

$$\frac{d^2y}{dx^2} = \frac{-1}{y^2}\left(\frac{dy}{dx}\right)^2 + \frac{1}{y}\frac{d^2y}{dx^2}$$

$$\frac{d^2y}{dx^2} - \frac{1}{y}\left(\frac{d^2y}{dx^2}\right) = \frac{-1}{y^2}\left(\frac{dy}{dx}\right)^2$$

$$\frac{d^2y}{dx^2}\left(1 - \frac{1}{y}\right) = \frac{-1}{y^2}\left(\frac{dy}{dx}\right)^2$$

$$\frac{d^2y}{dx^2}\left(\frac{y-1}{y}\right) = \frac{-1}{y^2}\left(\frac{dy}{dx}\right)^2$$

$$\frac{d^2y}{dx^2} = \frac{-1}{y}\left(\frac{dy}{dx}\right)^2$$

$$+\frac{d^2y}{dx^2}(1-y) = +\frac{1}{y}\left(\frac{dy}{dx}\right)^2$$

$$\frac{d^2y}{dx^2}(1-y) = \frac{1}{y}\left(\frac{dy}{dx}\right)^2$$

$$y''(1-y) = \frac{1}{y}(y')^2$$

Example

Lecture #03:-

iii) $x^2+y^2+2gx+2fy+c=0$

$$\frac{d}{dx}x^2+\frac{d}{dx}y^2+\frac{d}{dx}2gx+\frac{d}{dx}2fy+\frac{d}{dx}c=\frac{d}{dx}0$$

$$2x+2y\frac{dy}{dx}+2g+2f\frac{dy}{dx}+0=0$$

$$2x+2g+(2y+2f)\frac{dy}{dx}=0 \quad -(2$$

$$2x+2g=-(2y+2f)\frac{dy}{dx}.$$

Second Derivative.

$$\frac{d}{dx}2x+\frac{d}{dx}2g=-\left((2y+2f)\frac{d^2y}{dx^2}+\frac{dy}{dx}(0)\right)$$

$$2+0=-(2y+2f)\frac{d^2y}{dx^2}+$$

$$2\left[x+g+(y+f)\frac{dy}{dx}\right]=0$$

$$\frac{d}{dx}x+\frac{d}{dx}g+\frac{d}{dx}\left((y+f)\frac{dy}{dx}\right)=0$$

$$1+0+\left((y+f)y''+(y')^2\right)=0 \quad -(3$$

Third Derivative.

$$0+(y+f)\frac{d}{dx}\frac{d^2y}{dx^2}+y''\frac{dy}{dx}+2y'\cdot y''=0$$

$$y'''(y+f)+y'\cdot y''+2y'\cdot y''=0$$

$$y+f=\frac{-3y'y''}{y'''} \quad -(4$$

$$1+\left\{\left(\frac{-3y'y''}{y'''}\right)y''+(y')^2\right\}=0$$

$$1+(y')^2=\frac{3y'y''}{y'''}\cdot y''$$

$$y'''\left(1+(y')^2\right)=3y'\cdot(y'')^2$$

Example:- Find differential equation of all circles of radius.

$$(x-h)^2+(y-k)^2=a^2.$$

$$2(x-h)+2(y-k)\frac{dy}{dx}=0$$

$$(x-h)+1(y-k)\frac{dy}{dx}=0 \quad -i)$$

2nd Derivative.

$$1+(y-k)\frac{d^2y}{dx^2}+\left(\frac{dy}{dx}\right)^2=0$$

Now find $(y-k)$.

$$1+\left(\frac{dy}{dx}\right)^2=-(y-k)\frac{d^2y}{dx^2}$$

$$-1-\left(\frac{dy}{dx}\right)^2=(y-k)\left(\frac{d^2y}{dx^2}\right)$$

$$\frac{-1-(y')^2}{y''}=y-k \quad -3)$$

Put 3 in (i)

$$(x-h)+\cancel{\left(\frac{y-1-(y')^2}{y''}\right)y'=0}\left(\frac{-1-(y')^2}{y''}\right)(y')$$

$$x-h=\left(\frac{-1-(y')^2}{y''}\right)(y') \quad -(4)$$

Put 3 and 4 in (i)

$$\left(\frac{-1-(y')^2}{y''}\right)^2+\left(\left(\frac{-1-(y')^2}{y''}\right)(y')\right)^2=a^2.$$

$$\left(\frac{+1+(y')^2}{y''}\right)^2+\left(\frac{1+(y')^2(y')^2}{y''}\right)=a^2$$

$$\left(\frac{1+(y')^2}{y''}\right)\left(1+(y')^2\right)=a^2$$

$$\left(\frac{(1+(y')^2)^3}{(y'')^2}\right)=a^2$$

Lecture #04:-

Some mathematical models:- In science engineering economics, and even psychology, we often wish to describe or model the behaviour of some systems in mathematical terms. The description starts with

i) Identifying the variable, that are responsible for changing the system.

ii) A set of reasonable assumptions about the system.

These assumptions also include any empirical laws that are applicable to the system.

→ The mathematical model of the system, is in many cases is a differential equation or a system of differential equation.

→ The solution of model than gives the state of system.

In other words, for appropriate values of independent variables, the values of dependent variable(s) describe the system in the past, present and future.

Note :- "Modelling means turning a real situation into mathematical equations, so we can study it, explain it or predict what will happen".

Example #01: Falling Body

Throwing a ball upward.

Let $y(t)$ be the position of the ball.

→ Identify the forces acting on the ball.

$$F = -mg \quad -(i)$$

According to Newton's 2nd Law of motion

$$F = ma \rightarrow (ii)$$

$$ma = -mg$$

$$\frac{d^2y}{dt^2} + g = 0$$

$$a = \frac{dv}{dt}$$

$$v = \frac{dy}{dt}$$

$$= \frac{d}{dt}\left(\frac{dy}{dt}\right) = \frac{d^2y}{dt^2}$$

The mathematical description of a body falling vertically under the influence of gravity.

Example #2:- case study.

In a city with fixed population P peisons, the time rate of change of the number 'N' of the persons infected with the swine flu is proportional to the product of the number who have the disease and the number of those who do not.

→ P = Population

N = Number of infected persons.

$\frac{dN}{dt} = kN(P-N)$ } This D.E can be used to model the spread of virus through a community.

Example #3:- Insect population growth in a field.

A certain species of insects breeds in a crop field. Initially, food is abundant but as the insect population grows competition for food increases. The rate of growth of the insect population is proportional to the product of current insect population and the amount of food still available.

Let N(t) = no. of insects at time "t".

Let M represents maximum no. of insects the field can support.

$$\frac{dN}{dt} = KN(M-N)$$

★ Solution tells how insect population changes over time.

Assignment :-

i) Simple Pendulum's eq. of motion.

ii) Newton's Law of cooling. with real life example

iii) Case study of Covid-19.

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Lecture #5:-

Newton's Law of Cooling

It states that the rate at which body cools is directly proportional to the difference between the temperature of the body and the temperature of surrounding medium.

$$\frac{dT}{dt} = K(T - T_0)$$

∵ T = temperature of the body.

∵ T_0 = temperature of the surrounding medium.

Example # 4:- A veterinary doctor was examining a sick cat brought by a Pet lover. When it was brought to the hospital it was already dead. The pet lover wanted to find its time of death. He took the temperature of cat at 11:30 pm 94.6°F. He took the temperature again after one hour which was lower then 1st observation. It was 93.4°F. The room in which the cat was put is always at 70°F. The normal temperature of cat is 98.4°F. when it was alive.

The doctor estimates the time of death using Newton's law of cooling.

$$\frac{dT}{dt} = K(T - T_s), \quad T(0) = 94.6, \quad T(1) = 93.4$$

(Initial value problem; t = 1 hour)

First Order Differential Equation:

1) Methods of Separation of variables:- The differential equation of the type $F(x)G(y)dx + f(x)g(y)dy = 0$ is called separable equation.

$$\Rightarrow F(x)G(y)dx = -f(x)g(y)dy$$

2) $$\Rightarrow \int \frac{F(x)}{f(x)}dx = -\int \frac{g(y)}{G(y)}dy$$

Example # 01: $\frac{dy}{dx} = \frac{x^2}{y(1+x^3)}$

$$y\,dy = \frac{x^2}{1+x^3}dx \rightarrow \text{Separable D.E.}$$

Integral D.E:

$$\int y\,dy = \frac{1}{3}\int \frac{3x^2}{1+x^3}dx$$

$$\frac{y^2}{2} = \frac{1}{3}\ln|1+x^3| + \ln c. \qquad \because \ln a + \ln b = \ln ab, \quad a\ln b = \ln b^a$$

$$\frac{y^2}{2} = \frac{1}{3}\ln|1+x^3|^{1/3} + \ln c$$

$$\frac{y^2}{2} = \ln|c(1+x^3)^{1/3}|$$

$$e^{y^2/2} = c(1+x^3)^{1/3}$$

Example #02: $\frac{dy}{dx} + y^2 \sin x = 0$

on Separating variables

$$\frac{dy}{dx} = -y^2 \sin x$$

$$\int \frac{dy}{y^2} = -\int \sin x \, dx$$

$$\int y^{-2} dy = -\int \sin x \, dx$$

$$\frac{y^{-2+1}}{-2+1} = -(-\cos x) + c$$

$$\frac{-1}{y} = \cos x + c$$

Example #03 $\frac{dy}{dx} = 1 + x + y^2 + xy^2$

on separating variables.

$$\frac{dy}{dx} = (1+x) + y^2(1+x)$$

$$\frac{dy}{dx} = (1+x)(1+y^2)$$

$$\int \frac{dy}{1+y^2} = \int x \, dx + \int 1 \, dx$$

$$\tan^{-1}(y) = x + \frac{x^2}{2} + c$$

Example #04:- $\frac{dy}{dx} = 2x^2 + y - x^2 y + xy - 2x - 2$

on Separating variables.

$$\frac{dy}{dx} = 2(x^2 - x - 1) - y(x^2 - x - 1)$$

$$\frac{dy}{dx} = (2-y)(x^2 - x - 1)$$

$$\int \frac{dy}{2-y} = \int (x^2-x-1)dx$$

$$-\int \frac{(-1)dy}{(2-y)} = \int x^2 dx - \int x\,dx - \int 1\,dx$$

$$\Rightarrow -\ln|(2-y)| + \ln c = \frac{x^3}{3} - \frac{x^2}{2} - x$$

$$\ln\left|\frac{c}{2-y}\right| = \frac{x^3}{3} - \frac{x^2}{2} - x$$

Example #5:- $\text{Cosec}\,y\,dx + \sec x\,dy = 0$

on separating variables.

$$\text{Cosec}\,y\,dx = -\sec x\,dy$$

$$\frac{1}{\sec x}dx = -\frac{1}{\text{cosec}\,y}dy$$

$$\int \cos x\,dx = -\int \sin y\,dy$$

$$\sin x + c = -(-\cos y) + c$$

$$\sin x = \cos y + c$$

Example #6:- $y(1+x)dx + x(1+y)dy = 0$

$$y(1+x)dx = -x(1+y)dy$$

$$\int \frac{1+x}{x}dx = -\left(\frac{1+y}{y}\right)dy$$

$$\int \left(\frac{1}{x}+1\right)dx = -\int \left(\frac{1}{y}+1\right)dy$$

$$\int \frac{1}{x}dx + \int 1\,dx = -\int \frac{1}{y}dy + \int$$

$$\ln|x| + x + \ln c = -\ln|y| - y$$

$$x + y = -\ln|y| - \ln|x| - \ln c$$

$$\Rightarrow x + y = -[\ln xyc]$$

$$-(x+y) = \ln(xyc) \Rightarrow xyc = e^{-(x+y)} \Rightarrow xy = \frac{1}{c}e^{-(x+y)}$$

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Example #7: $(e^x+1)y\,dy = (y+1)e^x\,dx$

on separating variables.

$$\frac{y\,dy}{y+1} = \frac{e^x\,dx}{e^x+1}$$

$$\int \frac{y}{1+y}\,dy = \int \frac{e^x}{1+e^x}\,dx \Rightarrow \int \frac{y+1-1}{1+y}\,dy = \int \frac{e^x}{1+e^x}$$

$$\int \left(1-\frac{1}{y+1}\right)dy = \int \frac{e^x}{1+e^x}\,dx$$

$$\int 1\,dy - \int \frac{1}{y+1}\,dy = \int \frac{e^x}{1+e^x}\,dx$$

$$y - \ln|y+1| = \ln|1+e^x| + C$$

Swine flu spread:-

$$\frac{dN}{dt} = kN(P-N)$$

on Separating variables.

$$\int \frac{dN}{N(P-N)} = \int k\,dt = kt+C \quad -(A)$$

$$\frac{1}{N(P-N)} = \frac{A}{N} + \frac{B}{P-N}$$

$$\frac{1}{N(P-N)} = \frac{A(P-N)+BN}{N(P-N)} \quad -(B)$$

$$1 = A(P-N)+BN$$

Comparing coefficients

N^0: $1 = AP \Rightarrow A = 1/P$

N^1: $0 = B-A \Rightarrow A=B=1/P$

$$Eq\ (B) \Rightarrow \frac{1}{N(P-N)} = \frac{1}{PN} + \frac{1}{P(P-N)} = \frac{1}{P}\left[\frac{1}{N} + \frac{1}{P-N}\right]$$

$$Eq\ (A) \Rightarrow \int \frac{1}{P}\left[\frac{1}{N} + \frac{1}{P-N}\right] dN = Kt + C$$

$$\Rightarrow \frac{1}{P}\int \frac{1}{N}\, dN - \frac{1}{P}\int \frac{-1}{P-N}\, dN = Kt + C$$

$$\Rightarrow \frac{1}{P}\left[\ln|N| - \ln|P-N|\right] = Kt + C$$

$$\Rightarrow \frac{1}{P}\left[\ln\left(\frac{N}{P-N}\right)\right] = Kt + C$$

$$\Rightarrow \ln\left(\frac{N}{P-N}\right) = PKt + PC$$

Let $PC = C_1$

$$\Rightarrow \ln\left(\frac{N}{P-N}\right) = PKt + C_1$$

$$\text{or } \frac{N}{P-N} = e^{PKt + C_1}$$

$$\Rightarrow \frac{N}{P-N} = e^{PKt} \cdot e^{C_1} \quad \text{Let } e^{C_1} = C_2$$

$$\Rightarrow \frac{N}{P-N} = C_2 e^{PKt}$$

$$\Rightarrow N = (P-N)C_2 e^{PKt}$$

$$\Rightarrow N = PC_2 e^{PKt} - NC_2 e^{KPt}$$

$$\Rightarrow N(1 + C_2 e^{KPt}) = PC_2 e^{KPt}$$

$$\Rightarrow N(t) = \frac{PC_2 e^{KPt}}{1 + C_2 e^{KPt}}$$

Lecture # 6:-

Remaining Separable Differential Equation.

Example # 09: Solve $e^x\left(1+\frac{dy}{dx}\right)=xe^{-y}$

$$1+\frac{dy}{dx}=xe^{-y}\cdot e^{-x}=xe^{-(x+y)}$$

$$1+\frac{dy}{dx}=xe^{-(x+y)} \quad -3)$$

let $z=x+y \quad -1)$

$$\frac{dz}{dx}=1+\frac{dy}{dx} \quad -2)$$

using eq (1) and (2) in (3).

$$\frac{dz}{dx}=xe^{-z}$$

By separating variables.

$$\frac{dz}{e^{-z}}=x\,dx$$

$$\int e^z\,dz=\int x\,dx$$

$$e^z=\frac{x^2}{2}+C$$

$\because z=x+y$

$$e^{x+y}=\frac{x^2}{2}+C$$

Example # 10:- Solve the IVP ①

$$2(y-1)dy=(3x^2+4x+2)\,dx\ ;\ y(0)=-1$$

$$2\int(y-1)dy=\int(3x^2+4x+2)\,dx$$

$$2\left[\int y\,dy-\int dy\right]=3\int x^2dx+4\int x\,dx+2\int dx$$

$$\Rightarrow 2\left[\frac{y^2}{2} - y\right] = \frac{3x^3}{3} + \frac{4x^2}{2} + 2x + C$$

$$y^2 - 2y = x^3 + 2x^2 + 2x + C$$

Given $y(0) = -1$

at $x=0, y=-1$

$$\Rightarrow (-1)^2 - 2(-1) = (0)^3 + 2(0)^2 + 2(0) + C$$

$$\Rightarrow C = 3$$

Particular Solution is

$$y^2 - 2y = x^3 + 2x^2 + 2x + 3.$$

Example # 11: $(1 + 2y^2)dy = y\cos dx \; ; y(0) = 1$

By separation of variable.

$$\left(\frac{1+2y^2}{y}\right)dy = \cos x\, dx$$

Integrate both sides we get.

$$\int \left(\frac{1}{y} + 2y\right) dy = \int \cos x\, dx$$

$$\ln|y| + \frac{2y^2}{2} = \sin x + C$$

$$\ln|y| + y^2 = \sin x + C$$

General Solution.

Given $y(0) = 1$, at $x=0, y=1$

$$\ln|1| + 1 = \sin(0) + C$$

$$\Rightarrow C = 1$$

Hence: $\ln|y| + y^2 = \sin x + 1$ → Particular Solution.

Example # 12:- $\frac{dT}{dt} = K(T-T_0)$

By separating variables

$$\frac{1}{T-T_0}\,dT = K\,dt$$

$$\int \frac{1}{T-T_0}\,dT = \int K\,dt$$

$$\ln|T-T_0| = Kt + C$$

$$e^{\ln|T-T_0|} = e^{Kt+C}$$

$$T-T_0 = Ce^{Kt}$$

2) Method of Homogenous Differential Equation

Homogenous function:- If the function "f" = $f(x,y)$ has the property that $f(tx, ty) = t^n f(x,y)$

n is any real number.

f is said to be homogenous of degree n.

For example:- $f(x,y) = x^2 - 3xy + 5y^2$

$$f(tx, ty) = (tx)^2 - 3(tx)(ty) + 5(ty)^2$$

$$= t^2x^2 - 3(tx)(ty) + 5t^2y^2$$

$$= t^2(x^2 - 3xy + 5y^2)$$

$$= t^2 f(x,y)$$

$\Rightarrow$ $f(x,y)$ is homogenous fn of degree 2.

دن: تاریخ:

Lecture # 7

Homogenous Differential Equation: A differential equation of form $M(x,y)dy + N(x,y)dx = 0$ — (1 is said to be homogenous if $M(x,y)$ and $N(x,y)$ are homogenous function of same degree.

ie: $M(tx, ty) = t^n M(x,y)$

$N(tx, ty) = t^n N(x,y)$

For example :- $(x-y)dx + (x+y)dy = 0$

$M(x,y) = (x-y)$

$N(x,y) = (x+y)$

To solve homogenous differential equation.

Put $y = vx \rightarrow \frac{dy}{dx} = v + x\frac{dv}{dx}$ — (1

↳ (Separable Eqn) → Solve for y.

Example #01:- Solve $(x-y)dx + (x+y)dy = 0$ — (1

Eq 1) is a homogenous differential eq

To Solve it put $y = vx \Rightarrow \frac{dy}{dx} = v + x\frac{dv}{dx}$

$(x+y)dy = -(x-y)dx$

$\Rightarrow \frac{dy}{dx} = \frac{y-x}{y+x} \rightarrow$ (2

using eq (1) in (2 we get

$v + x\frac{dv}{dx} = \frac{vx-x}{vx+x} \Rightarrow v + x\frac{dv}{dx} = \frac{x(v-1)}{x(v+1)}$

$v + x\frac{dv}{dx} = \frac{v-1}{v+1} \Rightarrow x\frac{dv}{dx} = \frac{v-1}{v+1} - v$

$$\Rightarrow \frac{x\,dv}{dx} = \frac{v-1-v^2-v}{v+1} = \frac{-(1+v^2)}{v+1}$$

$$\Rightarrow x\frac{dv}{dx} = -\frac{(1+v^2)}{1+v}$$

on separating variables

$$\frac{v+1}{v^2+1}dv = \frac{-1}{x}dx$$

Integrate both sides.

$$\int \frac{v+1}{v^2+1}dv = \int \frac{-1}{x}dx$$

$$\int \left[\frac{v}{v^2+1} + \frac{1}{v^2+1}\right]dv = -\ln x + \ln c$$

$$\int \frac{v}{v^2+1}dv + \int \frac{1}{v^2+1}dv = -\ln x + \ln c$$

$$\frac{1}{2}\int \frac{2v}{v^2+1} + \tan^{-1}v = -\ln x + \ln c$$

$$\frac{1}{2}\ln|v^2+1| + \tan^{-1}v = -\ln x + \ln c$$

$$\ln|v^2+1|^{1/2} + \tan^{-1}v = \ln\left(\frac{c}{x}\right)$$

$$\tan^{-1}(v) = \ln\left(\frac{c}{x}\right) - \ln|v^2+1|^{1/2}$$

$$\tan^{-1}(v) = \ln\left|\frac{c}{x\,|v^2+1|^{1/2}}\right|$$

$$y = vx \Rightarrow v = y/x.$$

$$\tan^{-1}\left(\frac{y}{x}\right) = \ln\left|\frac{c}{x\left(\frac{y^2}{x^2}+1\right)^{1/2}}\right|$$

Homogenous DE of Degree 2.

Example #2:- Solve $(y^2+2xy)dx + x^2dy = 0$

$$(y^2+2xy)dx = -x^2dy$$

$$\frac{dy}{dx} = \frac{-x^2}{-x^2}\,\frac{y^2+2xy}{} - (1$$

دن: _____ تاریخ: _____

Put $y = vx \Rightarrow \frac{dy}{dx} = v + x\frac{dv}{dx}$ — (2

using eq (2 in (1

$$v + x\frac{dv}{dx} = -\left(\frac{v^2x^2 + 2vx^2}{x^2}\right)$$

$$v + x\frac{dv}{dx} = -\frac{x^2(v^2 + 2v)}{x^2}$$

$$v + x\frac{dv}{dx} = -(v^2 + 2v)$$

$$x\frac{dv}{dx} = -(v^2 + 2v) - v$$

$$x\frac{dv}{dx} = -v^2 - 2v - v$$

$$x\frac{dv}{dx} = -v^2 - 3v$$

$$x\frac{dv}{dx} = -v(v+3)$$

on separating variables: $\int \frac{1}{v(v+3)}dv = \int -\frac{1}{x}\,dv$ — A)

Consider, $\frac{1}{v(v+3)} = \frac{A}{v} + \frac{B}{v+3}$ — B)

$$1 = A(v+3) + Bv$$

$$1 = Av + 3A + Bv$$

comparing coefficients.

$$1 = 3A$$

$$1/3 = A. \quad -i)$$

$$0 = A + B$$

$$-1/3 = B$$

Eq B $\Rightarrow \frac{1}{V(V+3)} = \frac{1}{3V} - \frac{1}{3(V+3)} = \frac{1}{3}\left[\frac{1}{V} - \frac{1}{V+3}\right]$ →

using eq (C) in A.

$$\int \frac{1}{3}\left[\frac{1}{V} - \frac{1}{V+3}\right] dV = -\int \frac{1}{x} dx.$$

$$\frac{1}{3}\int \frac{1}{V} dV - \frac{1}{3}\int \frac{1}{V+3} dV = -\int \frac{1}{x} dx.$$

$$\frac{1}{3}\ln|V| - \frac{1}{3}\ln|V+3| = -\ln|x| + \ln c$$

$$\frac{1}{3}\ln\left|\frac{V}{V+3}\right| = -\ln\left|\frac{c}{x}\right| - 3\ln|x| + 3\ln|c|$$

$$\ln\left|\frac{V}{V+3}\right| = 3\ln\left|\frac{c}{x}\right| \quad \ln x^{-3} + \ln c^3$$

$$\ln\left|\frac{V}{V+3}\right| = \ln\left[c^3 x^{-3}\right]$$

$$\frac{V}{V+3} = \frac{c^3}{x^3}$$

Put $V = y/x$

$$\frac{y/x}{y/x + 3} = \frac{c^3}{x^3}$$

$$\frac{y}{y+3x} = \frac{c}{x^3}$$

Or

$$x^3 y = c(y+3)$$

تاریخ: _____ دن: _____

Example #03: $3x\cos\left(\frac{y}{x}\right)dy = \left[2x\sin\left(\frac{y}{x}\right) + 3y\cos\left(\frac{y}{x}\right)\right]dx$

put $y = vx \Rightarrow \frac{y}{x} = v$

$$\frac{dy}{dx} = v + x\frac{dv}{dx}$$

Eq (1) $\Rightarrow \frac{dy}{dx} = \frac{2x\sin\left(\frac{y}{x}\right) + 3y\cos\left(\frac{y}{x}\right)}{3x\cos\left(\frac{y}{x}\right)}$

$$v + x\frac{dv}{dx} = \frac{2x\sin(v) + 3vx\cos(v)}{3x\cos(v)}$$

$$v + x\frac{dv}{dx} = x\left(\frac{2\sin(v) + 3v\cos(v)}{3\cos v x}\right)$$

$$x\frac{dv}{dx} = \frac{2\sin(v) + 3v\cos(v)}{3\cos v} - v$$

$$x\frac{dv}{dx} = \frac{2\sin(v) + 3v\cos v - 3v\cos v}{3\cos(v)}$$

$$x\frac{dv}{dx} = \frac{2}{3}\frac{\sin(v)}{\cos(v)}$$

$$x\frac{dv}{dx} = \frac{2}{3}\tan(v)$$

on separating variables:

$$\int \frac{1}{\tan v}\,dv = \int \frac{2}{3}\frac{1}{x}\,dx$$

$$\int \frac{\cos v}{\sin v}\,dv = \frac{2}{3}\ln|x| + \ln c$$

$$\ln|\sin v| = \frac{2}{3}\ln|x| + \ln c$$

$$\ln|\sin v| = \ln|x|^{2/3} + \ln c$$

$$\ln|\sin v| = \ln(cx^{2/3})$$

$$\sin v = cx^{2/3}$$

put $v = y/x$

$$\sin\frac{y}{x} = cx^{2/3}$$

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Example #04: Solve $(x^2-3y^2)dx + 2xy\,dy = 0$

$$(x^2-3y^2)dx = -2xy\,dy$$

$$\frac{x^2-3y^2}{-2xy} = \frac{dy}{dx}$$

$$y = vx \Rightarrow \frac{dy}{dx} = \frac{dv}{dx} + v$$

$$\frac{x^2-3x^2v^2}{-2x\cdot x v} = \frac{dy}{dx}$$

$$\frac{x^2(1-3v^2)}{-2x^2v} = \frac{dy}{dx}$$

$$\frac{1-3v^2}{-2v} = \frac{dv}{dx} + v$$

$$\frac{1-3v^2}{-2v} - v = \frac{dv}{dx}$$

$$\frac{1-3v^2+2v^2}{-2v} = \frac{dv}{dx}$$

$$-\left(\frac{1-v^2}{2v}\right) = \frac{dv}{dx} \Rightarrow \frac{v^2-1}{2v} = \frac{dv}{dx}$$

$$\int dx = \int \frac{2v\,dv}{v^2-1}$$

$$\ln|x| + \ln c = \ln|v^2-1|$$

$$\ln cx = \ln|v^2-1|$$

$$cx = \frac{y^2}{x^2} - 1$$

$$cx^3 = y^2 - x^2$$

$$y^2 = x^2 + cx^3$$

$$cx^3 = y^2 - x^2$$

Example 5 :- Solve: The initial value problem

$$\frac{dy}{dx} = \frac{x+y}{x} \quad ; y(1)=1$$

let $y = vx$.

$$\frac{dy}{dx} = v + x\frac{dv}{dx}$$

$$v + x\frac{dv}{dx} = \frac{x+vx}{x}$$

$$v + x\frac{dv}{dx} = \frac{x(1+v)}{x}$$

$$x\frac{dv}{dx} = 1+v-v$$

$$x\frac{dv}{dx} = 1.$$

By separation of variables

$$\int dv = \int \frac{1}{x}\,dx.$$

$$v = \ln|x| + \ln|c| \quad \because \ln|c| = c_1$$

$$\frac{y}{x} = \ln|x| + c_1 \quad \text{— (1)}$$

→ General Solution.

$y(1)=1$

at $x=1$, $y=1$

$$\frac{1}{1} = \ln|1| = c$$

$$1 = c_1$$

put in (1)

$$\frac{y}{x} = \ln|x| + 1 \quad \text{or} \quad y = x(\ln|x|+1)$$

3) Equation Reducable to homogenous form:

The differential equation $(a_1x+b_1y+c_1)dx+(a_2x+b_2$ is not homogenous but it can be reduced to homogeno form if: $\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$ then make the transformation $x = X+h$, $y = Y+k$.

Example # 1:- Solve $\frac{dy}{dx} = \frac{x+3y-5}{x-y-1}$

It is not homogenous differential equation and $\frac{1}{1} \neq \frac{-3}{1}$ $(1 \neq -3)$.

To make eq(1) homogenous we use the following transformations: $x = X+h$, $y = Y+k$ } -(2)
$dx = dX$, $dy = dY$. }

Using eq (2) in eq (1) we get

$$\frac{dY}{dX} = \frac{X+h+3(Y+k)-5}{X+h-(Y+k)-1}$$

$$\frac{dY}{dX} = \frac{(X+3Y)+(h+3k-5)}{(X-Y)+(h-k-1)} \quad -(3$$

Put: $h+3k-5=0$ -C) $\Rightarrow$ $h+3k-5=0$

$h-k-1=0$ -D) $\pm h \mp k - 1 = 0$

$4k-4=0$

$k=1$.

using $k=1$ in eq (D)

$h-1-1=0$

$h=2$.

putting h and k in (3)

$$\frac{dY}{dX} = \frac{(X+3Y)+(2+3(1)-5)}{(X-Y)+(2-1-1)}$$

$$\frac{dY}{dX} = \frac{(X+3Y)+0}{(X-Y)+0} \quad -(E$$

→ Homogenous D.E.

Put $Y = VX \Rightarrow \frac{dY}{dX} = V + X\frac{dV}{dX}$

$$\text{eq } E \Rightarrow V + X\frac{dV}{dX} = \frac{X+3VX}{X-VX} = \frac{(1+3V)X}{(1-V)X}$$

$$V + X\frac{dV}{dX} = \frac{1+3V}{1-V}$$

$$X\frac{dV}{dX} = \frac{1+3V}{1-V} - V$$

$$X\frac{dV}{dX} = \frac{1+3V-V+V^2}{1-V}$$

$$X\frac{dV}{dX} = \frac{1+2V+V^2}{1-V} = \frac{(V+1)^2}{1-V}$$

on separating variables.

$$\int \frac{1-V}{(V+1)^2}\,dV = \int \frac{1}{X}\,dX$$

$$\int \frac{1}{(V+1)^2}\,dV - \int \frac{V}{(V+1)^2}\,dV = \int \frac{1}{X}\,dX$$

$$\int \frac{1}{(1+V)^2}\,dV - \int \frac{(V+1)-1}{(V+1)^2}\,dV = \int \frac{1}{x}\,dx$$

$$\int \frac{1}{(V+1)^2} - \int \frac{1}{1+V}\,dV + \int \frac{1}{(V+1)^2}\,dV = \int \frac{1}{x}\,dx$$

$$\Rightarrow 2\int \frac{1}{(1+V)^2}\,dV - \int \frac{1}{1+V}\,dV = \int \frac{1}{x}\,dx$$

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$$2\int (1+v)^{-2}dv - \int \frac{1}{(1+v)}dv = \int \frac{1}{x}dx$$

$$\frac{2(1+v)^{-2+1}}{-2+1} - \ln|1+v| = \ln|x| + \ln|c|$$

$$\frac{-2}{(v+1)} = \ln(x) + \ln|c| + \ln|1+v|$$

$$\frac{-2}{|v+1|} = \ln|x(1+v)c|$$

$$y = vx \Rightarrow v = \frac{y}{x}$$

$$\frac{-2}{\frac{y}{x}+1} = \ln|cx\left(1+\frac{y}{x}\right)|$$

$$\Rightarrow \frac{-2x}{x+y} = \ln|cx(x+y)|$$

using $X = x-2$ & $Y = y-1$ we get

$$\frac{-2(x-2)}{(x-2)+(y-1)} = \ln|c(x-2+y-1)|$$

$$\frac{-2(x-2)}{x+y-3} = \ln|c(x+y-3)|$$

Solution.

Homework question.

Q No $y+\sqrt{x^2+y^2}\,dx - x\,dy = 0$

$$y+\sqrt{x^2+y^2}\,dx = x\,dy$$

$$\frac{y+\sqrt{x^2+y^2}}{x} = \frac{dy}{dx}$$

let :- $y = vx$

$$\frac{dy}{dx} = v + x\frac{dv}{dx}$$

$$\Rightarrow \quad v+x\frac{dv}{dx} = \frac{vx+\sqrt{x^2+v^2x^2}}{x}$$

$$v+x\frac{dv}{dx} = \frac{vx+x\sqrt{1+v^2}}{x}$$

$$v+x\frac{dv}{dx} = \frac{x(v+\sqrt{1+v^2})}{x}$$

$$x\frac{dv}{dx} = v-v+\sqrt{1+v^2}$$

$$x\frac{dv}{dx} = \sqrt{1+v^2}$$

$$\int \frac{dv}{\sqrt{1+v^2}} = \int \frac{dx}{x}$$

$$\sinh^{-1}(v) = \ln|x| + c$$

$y(1) = 0$, $x = 1$, $y = 0$

$$v = \frac{y}{x} = \frac{0}{1} = 1$$

$$\sinh^{-1}(0) = \ln|1| + c$$

$$0 = c$$

$$\sinh^{-1}\left(\frac{y}{x}\right) = \ln|x|$$

:ن

$\ln|x+\sqrt{x^2+1}| = \ln(xc)$

$y(1)=0 \Rightarrow y=0, x=1$

$\frac{y}{x}+\sqrt{\frac{y^2}{x^2}+1} = xc$

$\frac{0}{1}+\sqrt{\frac{0}{1}+1} = 1c$

$0+\sqrt{0+1} = c$

$1 = c$

$\sin^{-1}h(v) = \ln|x|+1$

$Y(1) =$

Lecture #09:

Special case:- consider the non-homogenous equation

$(a_1x + b_1y + c_1)dx + (a_2x + b_2y + c_2)dy = 0$

and if $\frac{a_1}{a_2} = \frac{b_1}{b_2}$ then use transformation

$z = a_1x + b_1y$.

Example #01:- Solve $\frac{dy}{dx} = \frac{3x-4y-2}{3x-4y-3} \rightarrow (1)$

Now $\frac{a_1}{a_2} = \frac{b_1}{b_2} \Rightarrow \frac{3}{3} = \frac{-4}{-4} \quad 1 = 1.$

Let $z = 3x - 4y \rightarrow (2)$

$\frac{dz}{dx} = 3 - 4\frac{dy}{dx} \Rightarrow 3 - \frac{dz}{dx} = 4\frac{dy}{dx}.$

$\Rightarrow \frac{dy}{dx} = \frac{1}{4}\left(3 - \frac{dz}{dx}\right)$

using Eq's (2) and (3) in eq (1).

$\frac{1}{4}\left(3 - \frac{dz}{dx}\right) = \frac{z-2}{z-3}$

$3 - \frac{dz}{dx} = \frac{4z-8}{z-3}$

$\frac{dz}{dx} = 3 - \frac{4z-8}{z-3}$

$\Rightarrow \frac{dz}{dx} = \frac{3z-9-4z+8}{z-3}$

$\frac{dz}{dx} = \frac{-(z+1)}{z-3}$

This separable Differential Equation.

$\int \left(\frac{z-3}{z+1}\right)dz = -\int dx$

(ii)

$$\int \frac{(z-3+4)-4}{z+1}\,dz = -x+c$$

$$\Rightarrow \int \frac{(z+1)-4}{z+1}\,dz = -x+c$$

$$\int \left(\frac{z+1}{z+1}\,dz - \frac{4}{z+1}\,dz\right) = -x+c$$

$$= -x+c$$

$$\Rightarrow \int dz - 4\int \frac{1}{z+1}\,dz = -x+c$$

$$\Rightarrow z - 4\ln|z+1| = -x+c$$

put $z = 3x-4y$

$$(3x-4y) - 4\ln|(3x-4y)+1| = -x+c$$

4) Exact Differential Equation.

From calculus, we know if $z = f(x,y)$ is a function having continuous first-order partial derivative. then it's total differential is

$$dz = d(f(x,y))$$

$$dz = \frac{\partial f}{\partial x}\,dx + \frac{\partial f}{\partial y}\,dy \rightarrow (*)$$

Now if $f(x,y) = c \Rightarrow d(f(x,y)) = 0$

$$\frac{\partial f}{\partial x}\,dx + \frac{\partial f}{\partial y}\,dy = 0$$

↓ L.H.S ↓ R.H.S

For Example:- The equation $x^2y^3dx + x^3y^2dy = 0$ is exact since it is recognized that $d\left(\frac{1}{3}x^3y^3\right) = x^2y^3dx + x^3y^2dy$

$$d\left(\frac{1}{3}x^3y^3\right) = 0$$

$$\frac{1}{3}\cdot 3x^2y^3dx + \frac{1}{3}x^3y^2dy = 0$$

$$x^2y^3dx + x^3y^2dy = 0$$

$$d(f(x,y)) = \frac{\partial f}{\partial x}dx + \frac{\partial f}{\partial y}dy = 0$$

$$= \frac{1}{3}f[3x^2]$$

Eq (1) $\Rightarrow d\left(\frac{1}{3}x^3y^3\right) = 0 \Rightarrow \int d\left(\frac{1}{3}x^3y^3\right) = \int 0\,dx$

$$= \frac{1}{3}x^3y^3 = c \Rightarrow x^3y^3 = 3c$$

Definitions:- A D.E $M(x,y)dx + N(x,y)dy$ is an exact D.E if the expression on the left side is an exact differential.

Criterion for an Exact D.E:

Let $M(x,y)$ & $N(x,y)$ be continuous and have continuous first order partial derivative then an equation.

$M(x,y)dx + N(x,y)dy = 0$ is said to be an exact D.E if $M_y = N_x$

→ The solution of exact D.E is $f(x,y) = c$

Example #01: $2xy\,dx + (x^2-1)\,dy = 0 \rightarrow (1)$

$M(x,y) = 2xy$, $N(x,y) = x^2-1$

$\frac{\partial M}{\partial y} = My = 2x$, $\frac{\partial N}{\partial x} = Nx = 2x$

$My = Nx \Rightarrow$ Eq 1 is exact.

$\Rightarrow \exists$ a fun $f(x,y)$, such that $\frac{\partial f}{\partial x}dx + \frac{\partial f}{\partial y}dy = 0 \rightarrow (2)$

From eq's (1 and (2 $\frac{\partial f}{\partial x} = 2xy \rightarrow (3)$

$\frac{\partial f}{\partial y} = x^2-1$ (4)

Integrate Eq (3)

$\int \partial f = \int 2xy\,\partial x \Rightarrow f(x,y) = x^2y + g(y)$ (4)'

Diff Eq (4)' w.r.t 'y'

$\frac{\partial f}{\partial y} = x^2 + g'(y) - (5)$

Comparing (4) and (5)

$x^2-1 = x^2 + g'(y)$

$g'(y) = -1 \Rightarrow g(y) = -y$ (6)

using eq (6) in eq (4) we get,

$f(x,y) = x^2y - y$

The solution of eq (a) is $x^2y - y = c$

Example #2:- Solve $\underbrace{(3x^2+4xy)}_{M(x,y)}dx + \underbrace{(2x^2+2y)}_{N(x,y)}dy = 0$ —(1

$M_y = 4x$, $N_x = 4x$

$\Rightarrow M_y = N_x \Rightarrow$ Eq (1) is exact

$\Rightarrow \exists$ a fun $f(x,y)$ such that

$$\frac{\partial f}{\partial x}dx + \frac{\partial f}{\partial y}dy = 0 \rightarrow (2$$

$$\frac{\partial f}{\partial x} = 3x^2+4xy \ (3), \quad \frac{\partial f}{\partial y} = 2x^2+2y \ (4)$$

Integrating eq (3) $\int \partial f = \int (3x^2+4xy)\partial x$

$$f(x,y) = \frac{3x^3}{3} + \frac{4y x^2}{2} + g(y)$$

$$\Rightarrow f(x,y) = x^3 + 2x^2y + g(y) \rightarrow (5$$

To find: $g(y)$

Diff eq (5) w.r.t 'y' $\Rightarrow \frac{\partial f}{\partial y} = 2x^2 + g'(y)$ — (6)

Comparing eq (6) and (4) we get:

$2x^2 + 2y = 2x^2 + g'(y) \Rightarrow g'(y) = 2y$

$\Rightarrow g(y) = y^2 \rightarrow (7$

using eq(7) in eq.(5)

$$f(x,y) = x^3 + 2x^2y + y^2$$

The solution is

$$f(x,y) = c$$

$$x^3 + 2x^2y + y^2 = c$$

Integrating Factor [To convert non-exact D.E to exact D.E]

If a D.E $M(x,y)\,dx + N(x,y)\,dy = 0$ is not exact but when it is multiplied by a functn $\mu(x,y)$ & resulting equation $\mu(x,y)\,M(x,y)\,dx + \mu(x,y)\,N(x,y)\,dy = 0$ is exact, then $\mu(x,y)$ is called an integrating factor (I.F)

→ Method to find integrating factor Let $M(x,y)\,dx + N(x,y)\,dy = 0$ ---(1) is not exact ie $\neq M_y = N_x$

Then we have to find If by using the following rules.

Rule 1 :- If $\frac{M_y - N_x}{N} = p(x)$

then $I.F = e^{\int p(x)\,dx}$

Rule #2 - If $\frac{N_x - M_y}{M} = Q(y)$

then $I.F = e^{\int Q(y)\,dy}$

Rule #3 If $M(x,y)\,dx + N(x,y)\,dy = 0$ is not exact homogeneous and $xM+yN \neq 0$ then I.F $\frac{1}{xM+yN}$.

Rule # 4:- If $M(x,y)\,dx + N(x,y)\,dy = 0$ is not exact but can be expressed

$x f(xy)\,dx + y g(xy)\,dy = 0$

$xM - yN \neq 0$ then I.F $= \frac{1}{xM-yN}$

Example #01:- Solve $(xy^2+y)dx - x\,dy = 0$

$M(x,y) = xy^2+y$, $N(x,y) = -x$

$M_y = 2xy+1$, $N_x = -1$

$\Rightarrow M_y \neq N_x$

$\Rightarrow$ Eq (1) is not exact.

Now consider: $I.F = \frac{M_y - N_x}{N} = \frac{2xy+1+1}{-x}$

$= \frac{2xy+2}{-x} \neq P(x)$

Now: $I.F = \frac{N_x - M_y}{M} = \frac{-1-2xy-1}{xy^2+y}$

$I.F = \frac{-2xy-2}{xy^2+y} = \frac{-2(xy+1)}{y(xy+1)} = \frac{-2}{y} = Q(y)$

$I.F = e^{\int Q(y)\,dy} = e^{-\int \frac{2}{y}\,dy} = e^{-2\int \frac{1}{y}\,dy}$

$I.F = e^{-2\ln|y|} = e^{\ln y^{-2}} = y^{-2} = \frac{1}{y^2}$

$I.F = \frac{1}{y^2}$

Eq (1) $\Rightarrow \frac{1}{y^2}(xy^2+y)dx - \frac{x}{y^2}dy = 0$

$\Rightarrow \left(x+\frac{1}{y}\right)dx - \frac{x}{y^2}dy = 0$ — (2)

$M_y = \frac{-1}{y^2}$, $N_x = \frac{-1}{y^2}$

$\Rightarrow M_y = N_x$

Eq is exact D.E.

$\Rightarrow$ ∃ a fn $f(x,y)$ s.t

$\frac{\partial f}{\partial x}dx + \frac{\partial f}{\partial y}dy = 0$

$\Rightarrow \frac{\partial f}{\partial x} = x+\frac{1}{y}$ — (3) , $\frac{\partial f}{\partial y} = \frac{-x}{y^2}$ — (4)

Integrating eq (3)

$f(x,y) = \frac{x^2}{2} + \frac{x}{y} + g(y)$

Now $\frac{\partial f}{\partial y} = \frac{-x}{y^2} + g'(y)$

Comparing with eq (4) $\Rightarrow \frac{-x}{y^2} = \frac{-x}{y^2} + g'y$

$\Rightarrow g'(y) = 0$

$\Rightarrow g(y) = C_1$

$\Rightarrow f(x,y) = \frac{x^2}{2} + \frac{x}{y} + C_1$

The solution of exact DE is $f(x,y) = C$

$\Rightarrow \frac{x^2}{2} + \frac{x}{y} + C_1 = C$

$\Rightarrow \frac{x^2}{2} + \frac{x}{y} = C_2$

Say: $C - C_1 = C_2$

Linear Equation

The general form of first order and first degree linear equation.

$$\frac{dy}{dx} + p(x) \cdot y = Q(x) \quad - (1)$$

Solving a Linear First order Equation.

Step 1:- To solve a linear ~~First order equation~~ D.E first put it into the form (1), that is, make the coefficient of $\frac{dy}{dx}$ unity

Step #02:- Identify $P(x)$ and find the integrating factor $e^{\int P(x)dx}$.

Step #03:- Multiply the equation obtained in Step 1 by the integrating factor

$$e^{\int P(x)dx} \frac{dy}{dx} + e^{\int P(x)dx} \cdot p(x)\, y = Q(x)\, e^{\int P(x)dx}$$

Step #04:- The left side of equation in step #03 is the desirative of product of integrating factor and dependent variable y

$$\frac{d}{dx}\left[e^{\int P(x)dx} \cdot y \right] = Q(x) e^{\int P(x)dx}$$

Step #05:-

Integrate both sides of equation found in step #04.

Example 2: Solve $x\frac{dy}{dx} - 4y = x^6 e^x$ — 1)

Dividing b/s of eq (1) by 'x':

$$\frac{dy}{dx} - \frac{4}{x}y = x^5 e^x \quad , P(x) = -\frac{4}{x}$$

Integrating Factor :- $e^{\int P(x)dx} = e^{\int -\frac{4}{x}dx} = e^{-4\int \frac{1}{x}dx}$

$$= e^{-4\ln|x|} = e^{\ln x^{-4}} = x^{-4}$$

Multiplying Eq (2) by x^{-4}:

$$x^{-4}\frac{dy}{dx} - \frac{4}{x^5}y = x^5 \cdot x^{-4} e^x$$

$$\frac{d}{dx}\left[x^{-4}\cdot y\right] = xe^x$$

Integrating both sides.

$$\int d(x^{-4}y) = \int xe^x dx$$

$$x^{-4}y = xe^{-x} - \int e^x dx = xe^x - e^x + C$$

$$\Rightarrow x^{-4}y = xe^x - e^x + C$$

$$\text{or } \frac{y}{x^4} = xe^x - e^x + C$$

$$y = x^5 e^x - x^4 e^x + x^4 C \rightarrow \text{Required solution.}$$

Try $\left(\frac{dy}{dx} - 3y\right) = 0$ → homework.

Try & $\frac{dy}{dx} = \frac{1}{x+y^2}$; $y(-2) = 0$

Example #02:- Solve the initial value problem.

$$x\frac{dy}{dx} + y = 2x \quad , y(1) = 0 \longrightarrow$$

Solution:- dividing by 'x' on both sides.

$$\frac{dy}{dx} + \frac{y}{x} = 2 \rightarrow 1) \quad ; P(x) = \frac{1}{x}.$$

$$I.F :- \quad e^{\int P(x)dx} = e^{\int \frac{1}{x} dx} = e^{\ln x} = x.$$

Multiplying eq (1) by I.F

$$x\frac{dy}{dx} + y = 2x$$

$$\Rightarrow \frac{d}{dx}[xy] = 2x$$

Integrating both sides.

$$xy = x^2 + C \quad -2)$$

$$y = x + \frac{C}{x} \rightarrow \text{General Solution.}$$

using $y(1) = 0 \Rightarrow$ at $x = 1, y = 0$

$$0 = 1 + C \Rightarrow C = -1.$$

$$\Rightarrow y = x - \frac{1}{x} \rightarrow \text{Required Solution.}$$

Bernoulli Equation:- An equation of the form $\frac{dy}{dx} + P(x)y = Q(x)y^n$ is said to be Bernoulli's equation.

$$\frac{dy}{dx} + P(x)y^{\bullet} + 1) = Q(x)y^n$$

Dividing both sides by Bernoulli's equation.

$$\frac{1}{y^n}\frac{dy}{dx} + P(x)\frac{y}{y^n} = Q(x)\cdot\frac{y^n}{y^n}$$

$$y^{-n}\frac{dy}{dx} + p(x)y^{1-n} = Q(x) \rightarrow (2$$

Step#02. Let $y^{1-n} = t \Rightarrow (1-n)y^{-n}\frac{dy}{dx} = \frac{dt}{dx}$

$$y^{-n}\frac{dy}{dx} = \frac{1}{1-n}\frac{dt}{dx}$$

using eq (3) in eq (2) we get.

$$\frac{1}{1-n}\frac{dt}{dx} + p(x)t = Q(x)$$

$$\frac{dt}{dx} + p(x)(1-n)t = Q(x)(1-n) - 4)$$

This is linear differential equation of first order and first degree in dependent variable 't' and independent variable 'x'.

Step#03:- In this step replace value of t in general solution.

Example #01- Solve $x\frac{dy}{dx} + y = y^2 \ln x$ — (1)

Dividing eq (1) by 'x'.

$$\frac{dy}{dx} + \frac{y}{x} = \frac{y^2 \ln x}{x}$$

It is Bernoulli's equation.

Dividing eq (2) by y^2.

$$\frac{1}{y^2}\frac{dy}{dx} + \frac{1}{y^2}\cdot\frac{y}{x} = \frac{y^2 \ln x}{x}\cdot\frac{1}{y^2}$$

$$\frac{1}{y^2}\frac{dy}{dx} + \frac{1}{y}\left(\frac{1}{x}\right) = \frac{\ln x}{x} \quad -(3)$$

Let $\frac{1}{y} = t \Rightarrow -\frac{1}{y^2}\frac{dy}{dx} \Rightarrow \frac{1}{y^2}\frac{dy}{dx} = -\frac{dt}{dx}$ — 4

using eq (4) in eq (3) we get

$$-\frac{dt}{dx} + \frac{1}{x}\cdot t = \frac{\ln x}{x} \quad \text{or} \quad \frac{dt}{dx} - \frac{1}{x}t = -\frac{\ln x}{x} \quad -(4)'$$

It is linear differential equation in dependent variable 't' and independent variable x.

$$P(x) = -\frac{1}{x}$$

$$I.F = e^{\int P(x)dx} = e^{-\int \frac{1}{x}dx} = e^{-\ln x} = e^{\ln x^{-1}} = \frac{1}{x}$$

Multiplying eq (4)' by $I.F = \frac{1}{x}$

$$\frac{1}{x}\frac{dt}{dx} - \frac{1}{x^2}\cdot t = -\frac{\ln x}{x^2}$$

Integrating both sides.

$$\int d\left[\frac{t}{x}\right] = -\int \ln x \cdot x^{-2}\, dx$$

$$\Rightarrow \frac{t}{x} = -\left[\ln x \cdot \frac{x^{-1}}{-1} + \int x^{-1} \cdot \frac{1}{x}\, dx\right]$$

$$\frac{t}{x} = -\left[\frac{-\ln x}{x} + \int x^{-2}\, dx\right]$$

$$\frac{t}{x} = -\left[-\frac{\ln x}{x} - \frac{1}{x}\right] + c$$

$$\frac{t}{x} = \frac{\ln x}{x} + \frac{1}{x} + c$$

$$t = \ln x + 1 + cx$$

using $\frac{1}{y} = t$

we get

$$\frac{1}{y} = \ln x + 1 + cx$$

دن: تاریخ:

Orthogonal Trajectories:- (Application of Differential Equations)

When all the curves of one family of curves $G(x, y, c_1) = 0$ intersect orthogonally all the curves of another family $H(x, y, c_2) = 0$, then the families are said to be orthogonal trajectories of each other.

In other words, an orthogonal trajectory is any curve that intersects every curve of another family at right angles.

→ To find orthogonal trajectories we use D.E's

→ Firstly, we find the differential Equation.

$$\frac{dy}{dx} = f(x,y) \quad - (1$$

→ The D.E of second orthogonal family is then.

$$\frac{dy}{dx} = \frac{-1}{f(x,y)} \quad \rightarrow (2$$

Example #01:- Find the orthogonal trajectories of the family of ~~curves~~ rectangular hyperbolas.

$$y = \frac{c_1}{x} \quad \rightarrow (1$$

Diff eq (1) w.r.t. 'x'.

$$\frac{dy}{dx} = \frac{-c_1}{x^2}$$

From eq.1 $C_1 = xy$

$$\Rightarrow \frac{dy}{dx} = -\frac{xy}{x^2} = -\frac{y}{x} \Rightarrow \frac{dy}{dx} = \left(-\frac{y}{x}\right) \rightarrow f(x,y)$$

The differential equation of orthogonal trajectories is then:-

$$\frac{dy}{dx} = -\left[\frac{-y}{x}\right] = \frac{x}{y}$$

$$\Rightarrow \frac{dy}{dx} = \frac{x}{y}$$

$$\int y\,dy = \int x\,dx$$

$$\frac{y^2}{2} = \frac{x^2}{2} + C_1$$

$$y^2 = x^2 + 2C_1$$

$$\text{or } y^2 - x^2 = c \quad \because 2C_1 = c$$

Example #02:- Find the orthogonal trajectories of $y = \frac{C_1 x}{1+x}$

Sol :- Given:- $y = \frac{C_1 x}{1+x}$

Differentiate eq #1 w.r.t 'x'

$$\Rightarrow \frac{dy}{dx} = \frac{(1+x)C_1 - C_1 x(1)}{(1+x)^2} = \frac{C_1 + C_1 x - C_1 x}{(1+x)^2}$$

$$\frac{dy}{dx} = \frac{C_1}{(1+x^2)}$$

From eq (1)

$$y(1+x) = C_1 x$$

$$\Rightarrow C_1 = \frac{y}{x}(1+x)$$

$$\Rightarrow \frac{dy}{dx} = \frac{y(1+x)}{x(1+x)^2}$$

$$\frac{dy}{dx} = \frac{y}{x(1+x)}$$

The D.E of orthogonal trajectories is

$$\frac{dy}{dx} = -\frac{x(1+x)}{y}$$

$$\Rightarrow y\,dy = -x(1+x)\,dx$$

Integrating b.s.

$$\int y\,dy = -\int x\,dx - \int x^2\,dx$$

$$\frac{y^2}{2} = -\frac{x^2}{2} - \frac{x^3}{3} + C$$

$$= \frac{-3x^2 - 2x^3 + 6C}{6} \qquad \because 6C = C_2.$$

$$y^2 = 12\left(\frac{-3x^2 - 2x^3}{6} + C_2\right)$$

$$3y^2 + 3x^2 + 2x^3 = C_2$$